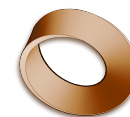


L0 – First Meeting and Coverage Audit

P2 Electricity • AQA Combined Science: Trilogy 8464



Physics 11 Q • Mr. Rawson • 14 students

11 Q Times	Wed 2 Sep, P3+P4 (Week B) P3 11:15–12:10 (55 min)	Next P4	Fri 4 Sep, P1 — 2 days 12:15–13:05 (50 min), into lunch
Spec To print	P2 — whole-unit audit 14 × course expectations (y11q-physics-gcse.pdf); the audit (p2-electricity-audit.pdf) split by page — p.1 × 14 for students, p.2 × 1 for you. Page 2 is the teacher master. It names a colleague. It is never issued to students and never published.	Lesson	L0 — precedes the sequence
Homework	None set — first meeting. [[ALEX: decide — whether to set anything before Fri 4 Sep.]]		

By the end of the double:

- every student is greeted by name, name-tagged, and joined to Google Classroom;
- the course expectations have been delivered and the sheet is in their hands;
- an honest coverage picture of P2 exists, tallied against Robert's record;
- you have a first read on which “secure” electricity actually is secure.

What this audit is for — and how 11 Q differs from 11 R. 11 R had no inherited record, so their audit *established* coverage. 11 Q have one: Robert recorded 92 statement-level entries. So today's audit **tests the record against the students' own sense of it**. A row Robert marked **Yes** where several of the room tick *shaky* is the most valuable thing this lesson can surface — it tells you the record overstates what you can build on. Watch for it row by row as you tally.

Timing — P3 (11:15–12:10)

Time	Phase	What happens
0–7	At the door	Greet each student at the door; name tags at the desks. First impressions run one way.
7–10	Who I am	Brief. The expectations sheet carries the detail — do not lecture it twice.
10–22	Expectations	Walk the sheet (14 copies). Same delivery as 11 R on 28 Aug.
22–30	Google Classroom	Everyone joined before moving on. [[ALEX: decide — join code, and whether devices are to hand in PH1.]]
30–33	Frame the audit	Say it out loud: <i>not a test, not marked</i> . Guessing high costs them lessons they needed. One tick per row.
33–48	The audit	Page 1 only, worked individually. Circulate; the free-text lines at the bottom (“anything not on this list” / “worries you most”) are often the best data on the sheet.
48–55	Collect + buffer	Collect all 14. Early finishers to PhET (below). Quick first scan while they pack for changeover.

Timing — P4 (12:15–13:05)

Time	Phase	What happens
0–8	Tally	While they settle: rapid tally against Robert’s record (table overleaf). You are hunting disagreements , not totals.
8–28	Retrieval	Quickfire on the claimed- Yes ground: $Q = It$, $V = IR$, series rules, $P = VI$. Whiteboards or cold-call. Hinge question: “ <i>Same current everywhere in a series loop — true or false, and why?</i> ” — that statement was missed in Robert’s run and underpins all of P2.2. Pitch the depth at what the tally showed. [[ALEX: decide — live, from the P3 audit result.]]
28–36	The map	Show them the honest position: what is done, what is left (parallel circuits, <i>all</i> calculation practice, all of domestic electricity, the National Grid). Frame it as “here is our run-in”, not as a deficit.
36–44	The year ahead	P2 finishes first, then the remaining units, then exams in summer 2027. [[ALEX: decide — mock dates and the whole-year sequence; do not improvise these in front of them.]]
44–50	PhET	Circuit Construction Kit: DC — build a series circuit, then a parallel one. Browser, no login. Doubles as the flexible tail into lunch.

Robert's recorded coverage for 11 Q (from the audit's page 2)

Ref	Statement	Record
P2.1.1	Circuit symbols	Part
P2.1.2	Charge and current, $Q = It$	Yes
P2.1.3	Current, p.d., resistance, $V = IR$	Yes
RP15	<i>Practical</i> : resistance factors	Part
P2.1.4	Resistors, non-ohmic components	No
RP16	<i>Practical</i> : I – V characteristics	No
P2.2a	Series circuits	Yes
P2.2b	Parallel circuits	No
P2.3.1	d.c. and a.c.	No
P2.3.2	Mains electricity and safety	No
P2.4.1	Power, $P = VI$, $P = I^2R$	Yes
P2.4.2	Energy transfers in appliances	Yes
P2.4.3	National Grid, transformers [HT]	No

Four specifics worth probing, from that page

- **RP15 is half done** — resistors in series and parallel were investigated; *wire length was not*. Ask specifically.
- **P2.2 splits cleanly** — series taught, parallel not, and **no calculations by either route**. Calculation is the gap, not the concept — a student can honestly tick “fine” on series while never having computed anything.
- **One statement missed in P2.1.2** — “a current has the same value at any point in a single closed loop”. Small, but it underpins everything in P2.2. It is the hinge question in P4.
- **All of P2.3 is outstanding** — 13 statements, and cheap to teach.

Common misconceptions

For the retrieval quickfire — the standard electricity errors to listen for:

- **“Current gets used up”** around the circuit. The missed loop statement is the direct antidote — if this surfaces, it confirms the gap.
- **Current and p.d. conflated** — “the voltage flows through”.
- **“Parallel branches always share the current equally.”** Only with identical branches.
- **$V = IR$ recalled but not usable** — can quote it, cannot pick out which quantity is which in a worded question. This is exactly the calculation gap the record predicts.

Differentiation

Support: on the audit, reassure that “New” is a fine answer — the sheet says so, say it again in person. On PhET, start everyone from the same two-bulb series build before letting them roam.

Extend: PhET — add an ammeter and a voltmeter, predict the readings before placing them, then explain any surprise. That is RP16 territory, which this class has never met.

After the lesson

- Flip this entry to **Taught** in the 11 Q Class Log.
- **Record every row where the room disagreed with Robert's record** — especially Yes-but-shaky. That list, not the record, is what the P2 run-in gets pitched at.
- [[ALEX: decide — where the 11 Q tally lives. Page 2's tally grid is headed "11R tally (of 10)" — the wrong class and the wrong roll for this lesson. Relabel it by hand, tally on scratch paper, or rebuild page 2 for 11 Q. The spreadsheet already holds Robert's column, so a second column beside it is another option. Do not edit the audit file without deciding.]]
- Read the two free-text answers on all 14 sheets — "worries you most" in particular.
- Note how far the PhET task got — it is the natural warm-up for Fri 4 Sep, P1.
- Fri 4 Sep is **2 days away**: pitch it from today's tally before the audit goes cold.